

10,000



Account A

Account B

UPDATE account A ....

UPDATE account B ...

ACID

Transactions

ACID

Atomocitiy :

Consistency

Isolation

Durability



Account c

Rs 1000

T1 : withdraw 2000

T2 : withdraw 3000

T1 : 4000

T2 : 3000

## Physical Transaction

```
@Transactional
public void transfer(Long fromId, Long toId, BigDecimal amount) {
    // transaction begins before this method executes

    debitAccount(fromId, amount);
    creditAccount(toId, amount);
    recordTransfer(fromId, toId, amount);

    // transaction commits after successful method completion
}
```

BEGIN  
COMMIT  
ROLLBACK

Spring JPA

JPA / Hibernate

JDBC

```
beginTransaction();
try {
    op1
    op2
    commitTransaction();
}
catch(Excetion e) {
    rollBackTransaction();
}
```

## Account

| id | name | balance |
|----|------|---------|
|    |      |         |

## TransferRecord

| id | fromAccId | toAccId | amount | createdAt |
|----|-----------|---------|--------|-----------|
|    |           |         |        |           |

Account#1

Account#2

TransferRecord#1

PC